Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1 Loading tidyverse, here, cowplot packages, and veryifying home directory
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
library(here)
```

```
## here() starts at /home/guest/ciraortizenviron872
```

```r
library(ggthemes)
```

```
##
## Attaching package: 'ggthemes'
##
## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```r
here()
```

```
## [1] "/home/guest/ciraortizenviron872"
```

```r
#2 Loading NTL LTR data
PeterPaulChem<-read.csv(here
        ("./Data/Processed/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
         stringsAsFactors = T)

PeterPaulPhys<-read.csv(here
        ("./Data/Processed/Processed_KEY/NTL-LTER_Lake_ChemistryPhysics_PeterPaul_Processed.csv"),
         stringsAsFactors = T)

Niwotlitter<-read.csv(here("Data/Processed/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),
          stringsAsFactors = T)
class(Niwotlitter$collectDate)      #as a factor
```

```
## [1] "factor"
```

```r
Niwotlitter$collectDate<-ymd(Niwotlitter$collectDate)     #changing to date
PeterPaulChem$sampledate<-ymd(PeterPaulChem$sampledate)
PeterPaulPhys$sampledate<-ymd(PeterPaulPhys$sampledate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```r
#3 - Customizing my own default theme

my_theme <- theme_base() +
  theme(
    #customize plot background
    plot.background = element_rect(fill = "pink", color = "black"),
    #customize plot title
    plot.title = element_text(size=16, face="bold",color="purple"),
    #customize axis labels
    axis.title = element_text(size=12, color = "black"),
    #customize ticks/gridlines
    panel.grid.major= element_line(color="darkgrey", linetype = "dashed"),
    panel.grid.minor = element_line(color="darkgrey", linetype ="dashed"),
    panel.background=element_rect(fill="white"),
    #customize legend
    legend.background = element_rect(fill="white", color="purple"),
    legend.position="right"
  )

#set my theme as default
theme_set(my_theme)
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

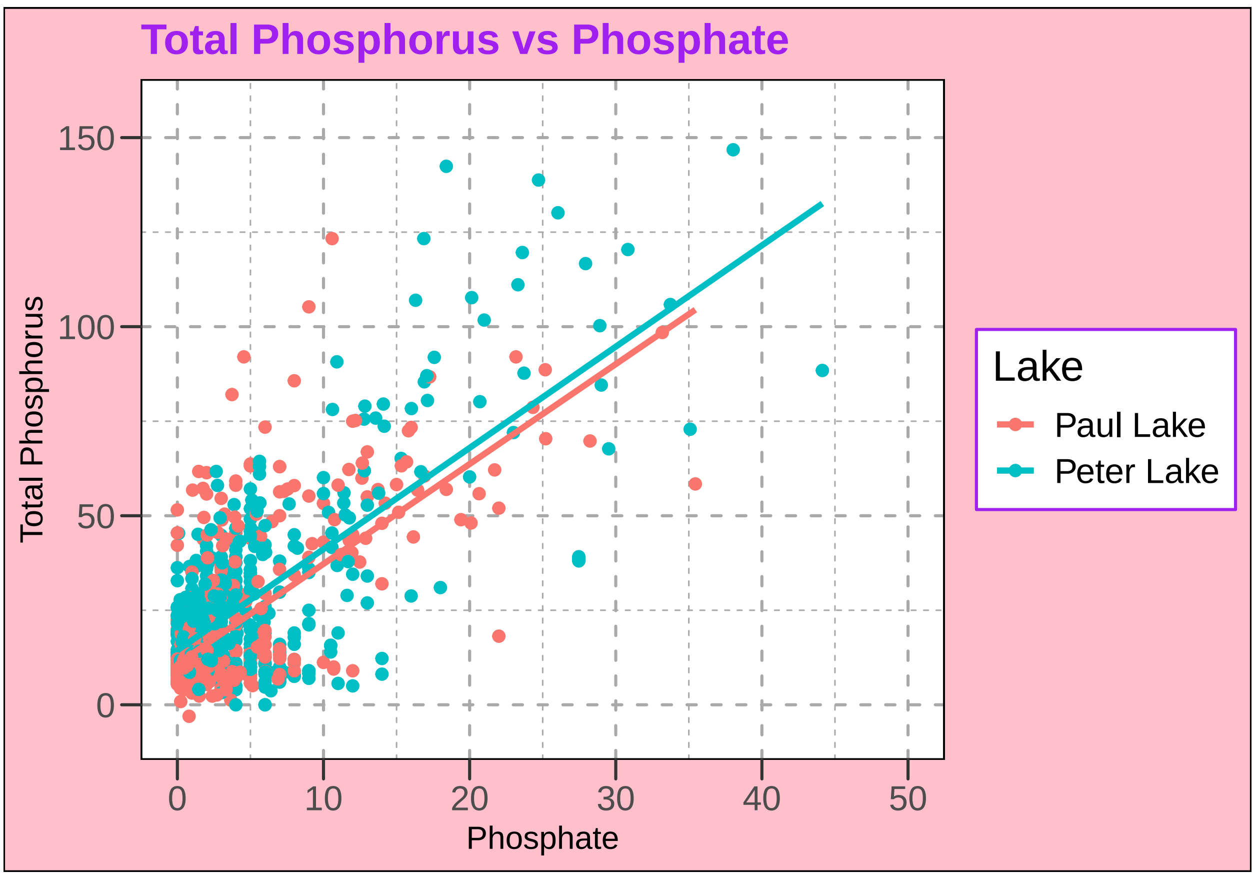
```r
#4
library(ggplot2)
plotex4<- PeterPaulChem %>%
  ggplot(aes(x=po4,
             y=tp_ug,
             color=lakename,
             ))+
           geom_point()+
  #adding lines of best fit
  geom_smooth(method=lm, se=FALSE)+
           labs(
             title="Total Phosphorus vs Phosphate",
             x="Phosphate",
             y="Total Phosphorus",
             color="Lake",
```

```r
          )+
  #setting limits on axes to remove outliers
  xlim(0,50)
ylim(0,150)
```

```
## <ScaleContinuousPosition>
##  Range:
##  Limits:    0 --  150
```

```r
plotex4
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 21947 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21947 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

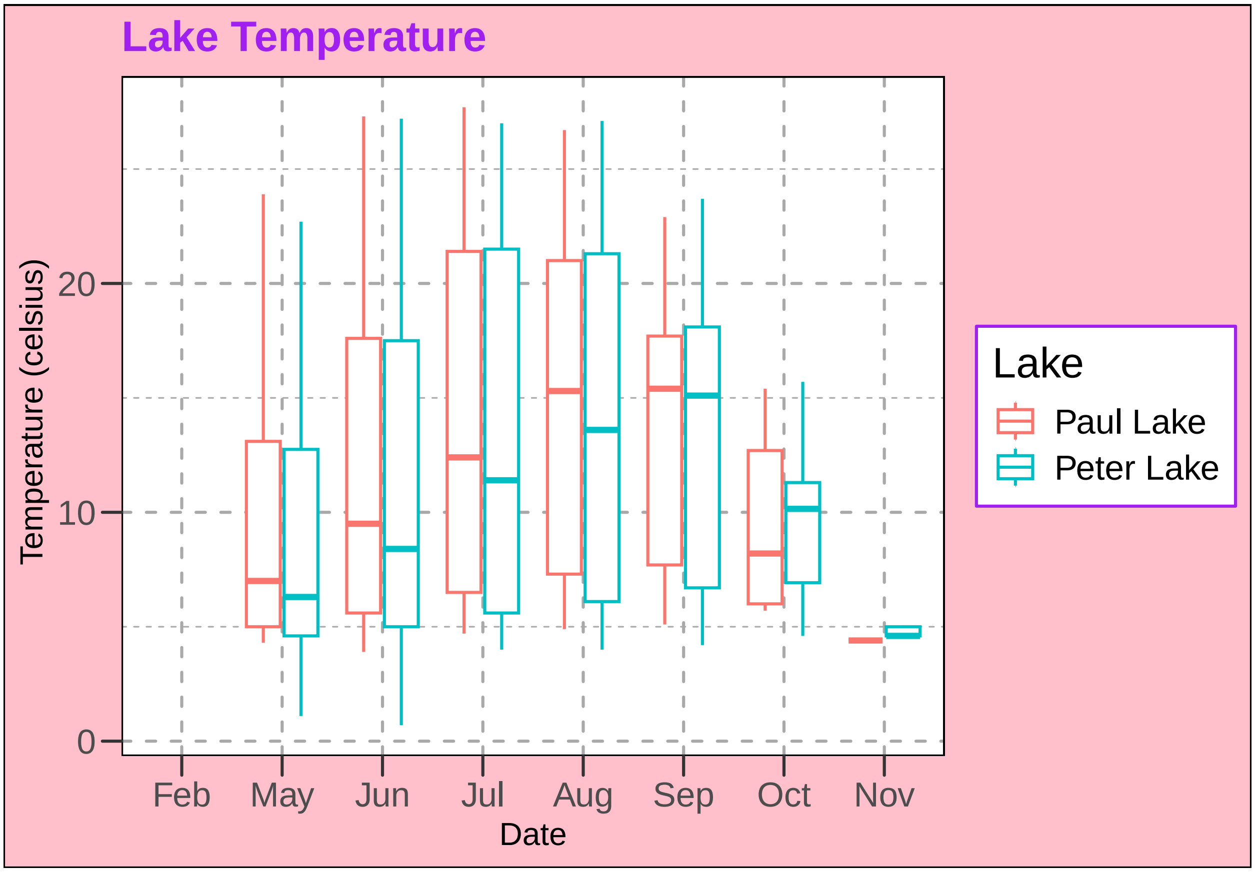
Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
#changing month from integer to factor
unique(PeterPaulChem$month) #checking what values I have in the month column
```

```
## [1]  5  6  7  8  9 10 11  2
```

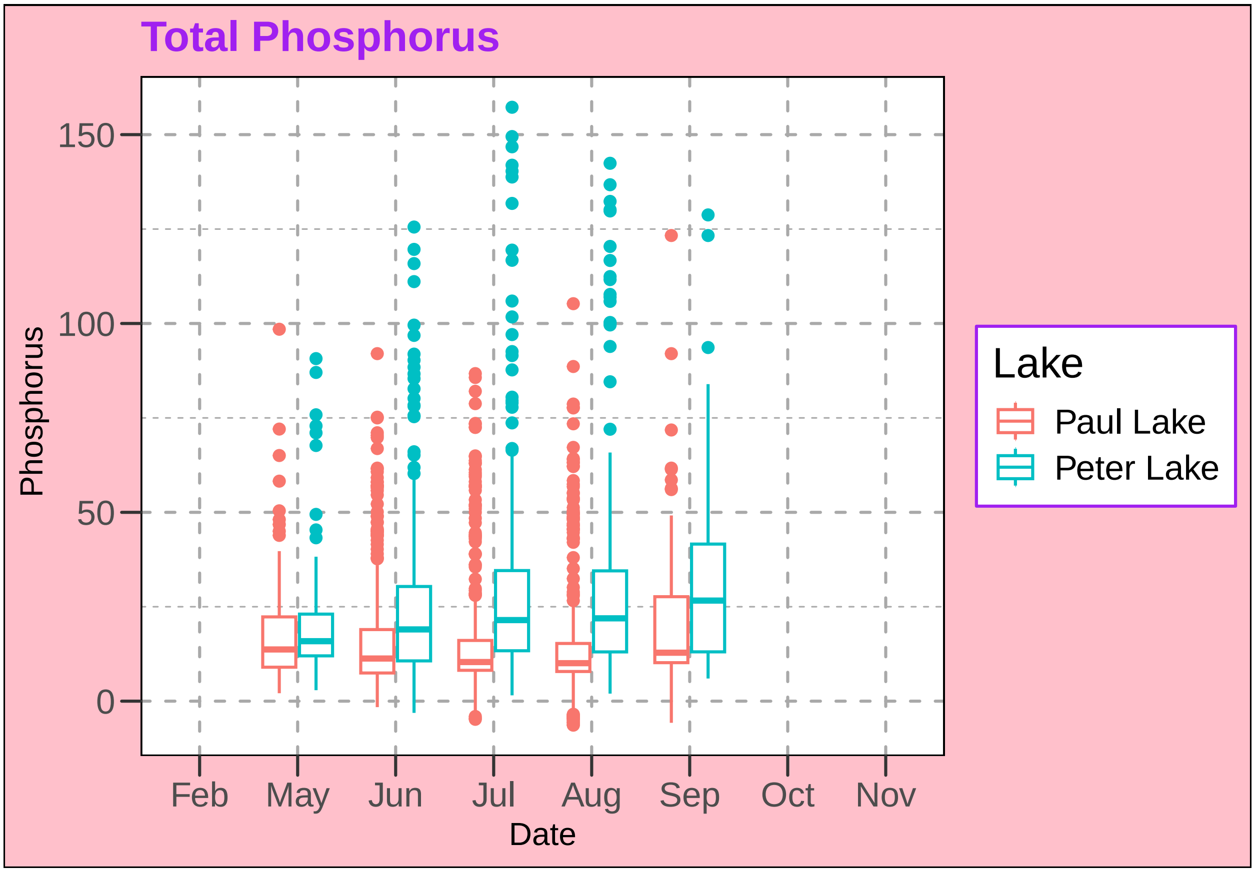
```
PeterPaulChem$month<-factor(
  PeterPaulChem$month,
  levels = c(2,5,6,7,8,9,10,11), #these are the integer months I currently have
  labels = c("Feb", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov")  #changing their labels
  #to abbreviated names
)

#a. Temperature boxplot
tempbox<-PeterPaulChem %>%
  ggplot(aes(
    x = month,
    y = temperature_C,
    color = lakename
  ))+
  geom_boxplot()+
    labs(title = "Lake Temperature",
       x = "Date",
       y = " Temperature (celsius)",
       color = "Lake")

tempbox
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



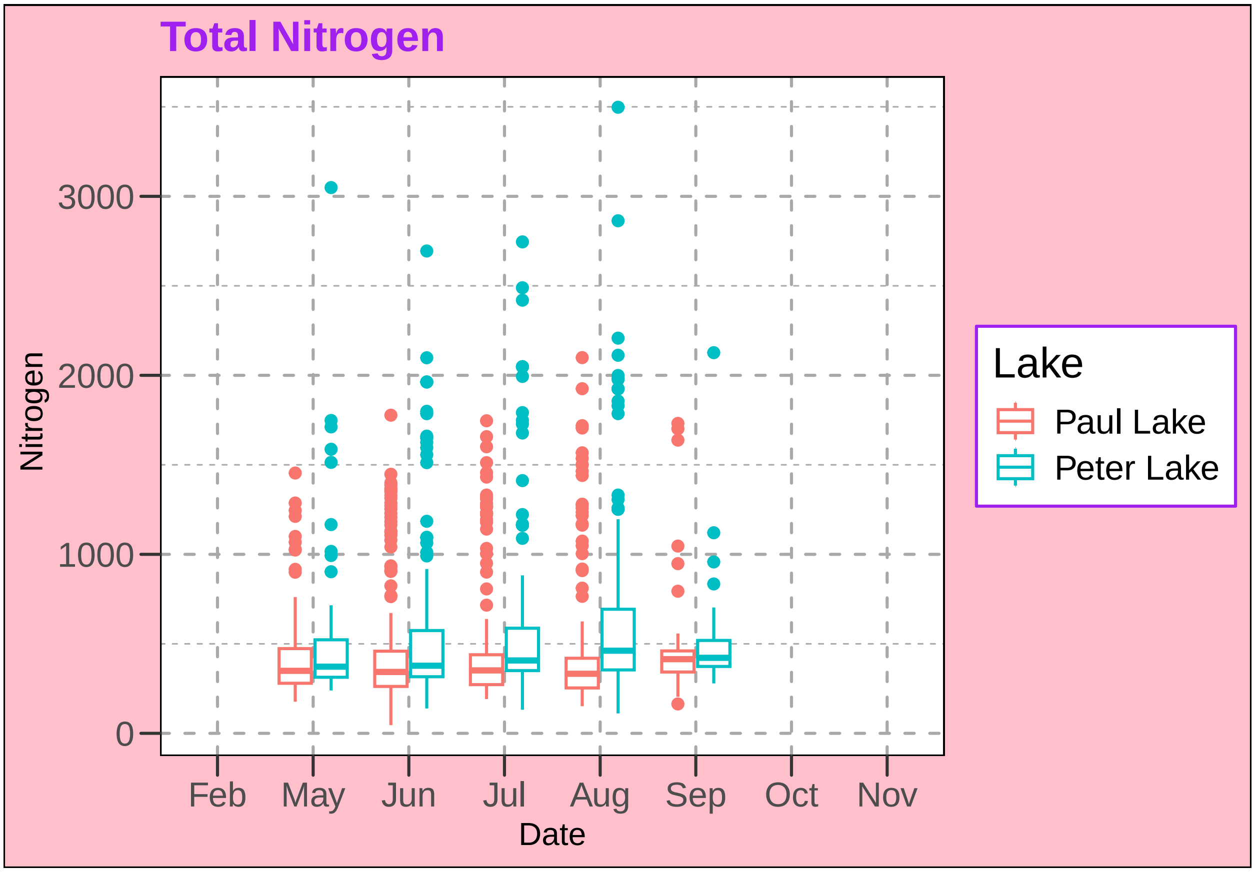

```
#b. TP boxplot
tpbox<-PeterPaulChem %>%
  ggplot(aes(
    x = month,
    y = tp_ug,
    color = lakename
  ))+
  geom_boxplot()+
      labs(title = "Total Phosphorus",
       x = "Date",
       y = " Phosphorus",
       color = "Lake")

tpbox
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
#c. TN boxplot
tnbox<-PeterPaulChem %>%
  ggplot(aes(
    x = month,
    y = tn_ug,
    color = lakename
  ))+
  geom_boxplot()+
        labs(title = "Total Nitrogen",
       x = "Date",
       y = " Nitrogen",
       color = "Lake")

tnbox
```

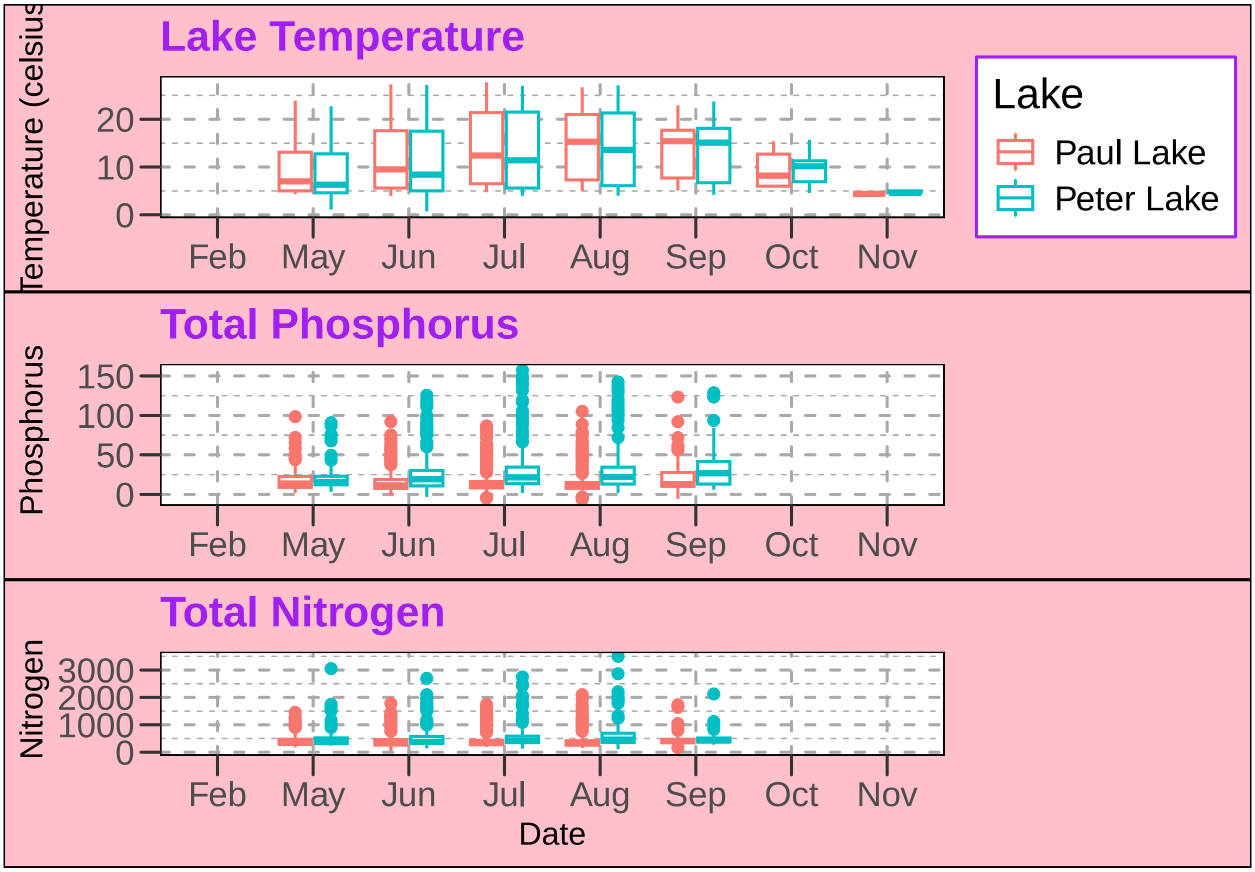
```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```




```
#d. Cowplot combining a,b,c
#making sure

combinedplot<-plot_grid(
  tempbox + theme(axis.title.x = element_blank()),    #this will be the top plot,
  #so leaving legend, removing x axis label
  tpbox + theme(axis.title.x = element_blank(), legend.position ="none"), #removed x axis label and leg
  tnbox + theme(legend.position = "none"),   #removed legend,
  #kept x axis label since it's on the bottom
  ncol =1,        #stacking plots vertically in one column
  align ="v",     #aligns same x-axis position
  rel_heights = c(1,1,1) #ensuring all plots are the same height
)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 20729 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 21583 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
combinedplot
```



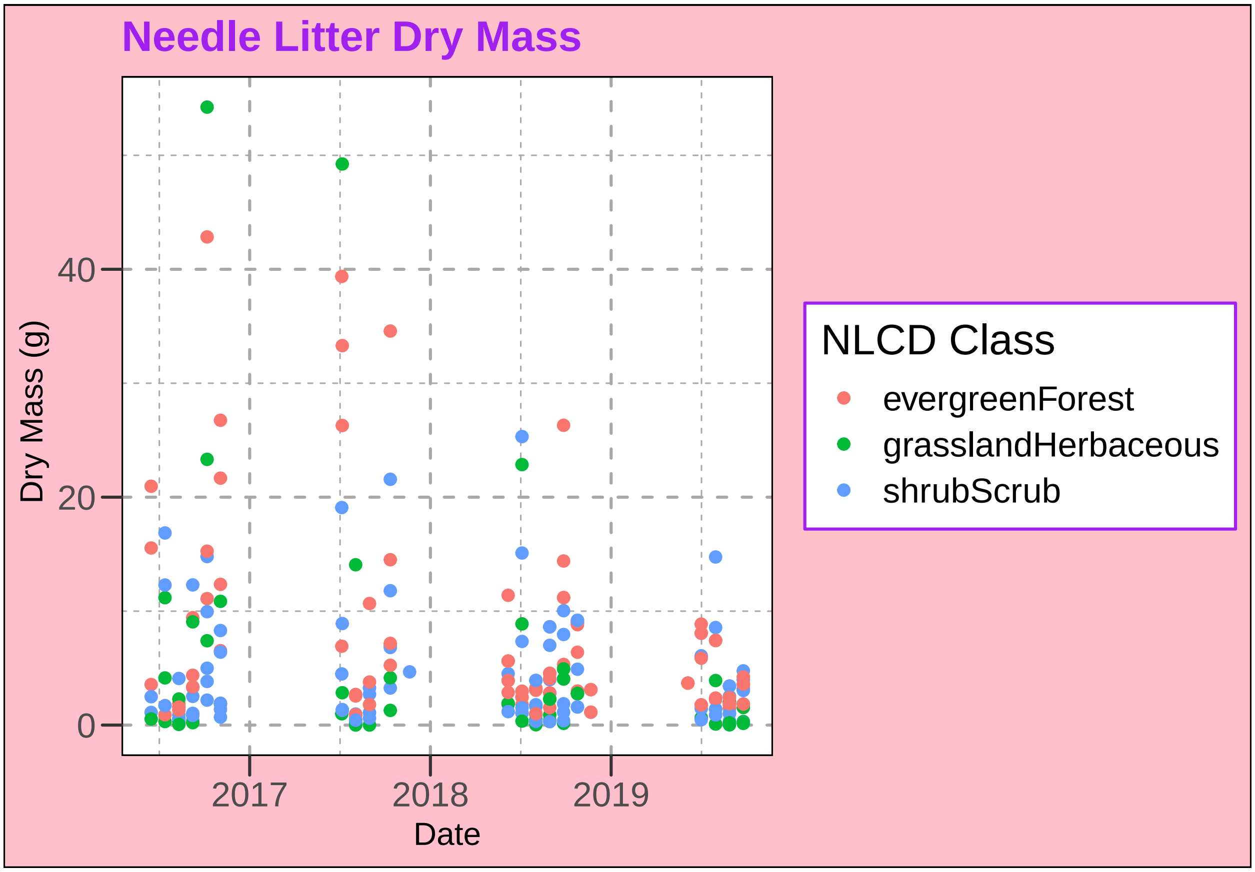
Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: Temperature follows a relatively normal seasonal patter, with the mean temperature increasing steadily over the summer. There are also no extreme outliers in the temperature plot, but there is more variability compared to the other variables. The phosphorus data is relatively concentrated around 20-25, with more cases of outliers, but no strong seasonsal rise or fall except for a small increase in July and August. Peter Lake consistently shows higher levels of phosphorus as well. Nitrogen levels are also relatively concentrated around 500, with Peter Lake again averaging at a higher rate, and having more outliers regardless of the season.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

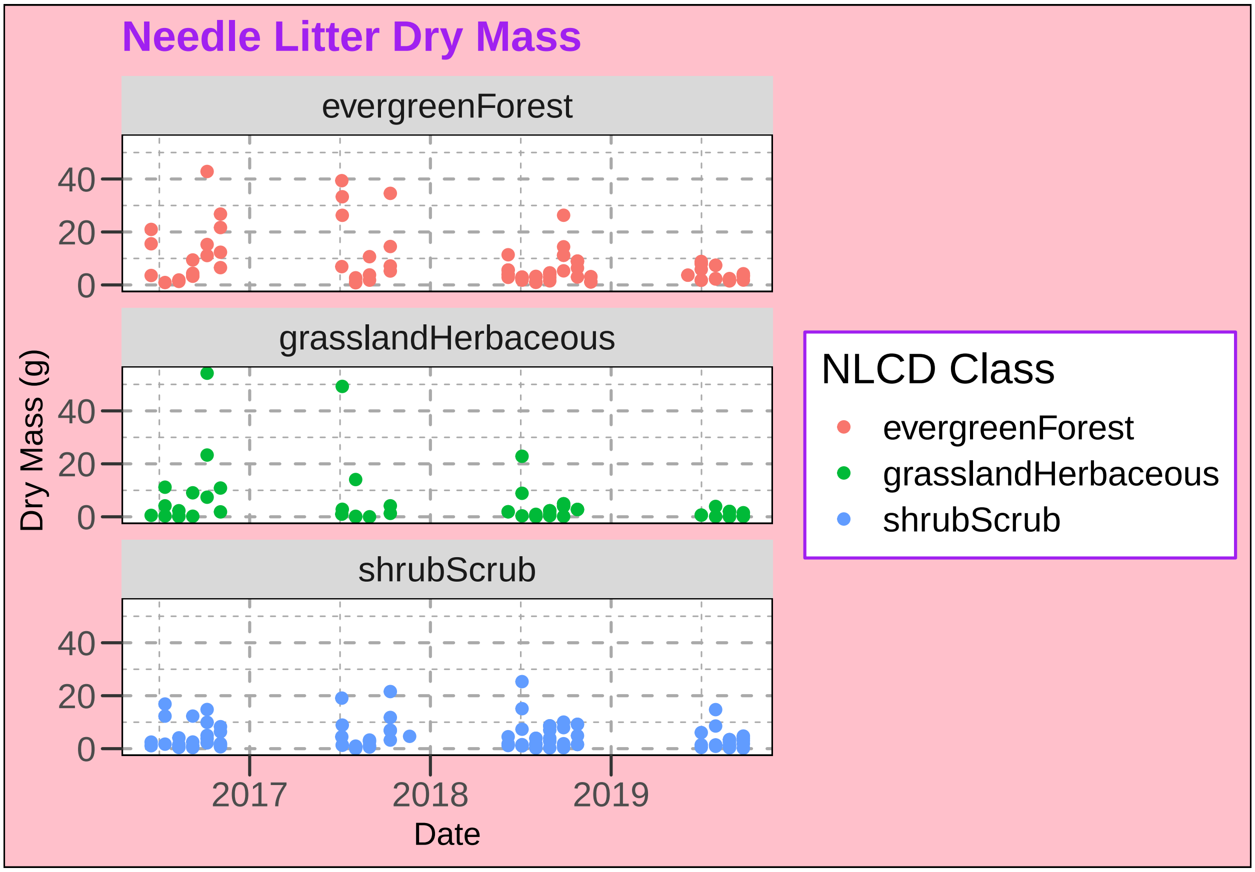
```
#6
needles<-Niwotlitter %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(aes(
    x = collectDate,
    y = dryMass,
```

```
    color = nlcdClass
  ))+
  geom_point()+
  labs(title = "Needle Litter Dry Mass",
       x = "Date",
       y = " Dry Mass (g)",
       color = "NLCD Class")

needles
```




```
#7
needles2<-Niwotlitter %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(aes(
    x = collectDate,
    y = dryMass,
    color = nlcdClass
  ))+
  geom_point()+
  facet_wrap(facets =vars(nlcdClass), #identifying which variable I want to be faceted
             nrow=3)+ #number of plots = number of land types
  labs(title = "Needle Litter Dry Mass",
       x = "Date",
       y = " Dry Mass (g)",
       color = "NLCD Class")
```

```
needles2
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer: The faceted plot, #7, is more effective since you can clearly delineate between the different kinds of land. The first plot, is very busy, with many of the points all stacked on top of each other making it hard to identify which type of land the litter is associated with.